Task 4 – Capstone Feature

Jetpack Compose Themed UI Implementation

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Technology: Jetpack Compose + Material Design 3
Applications Developed

- Superheroes Application
- 30 Days of Fitness Application

1 Feature Overview

This task demonstrates the implementation of modern Android user interfaces using Jetpack Compose and Material Design 3.

Two applications were developed to demonstrate UI layouts, scrollable lists, themed components, and responsive design.

The **Superheroes Application** displays a list of heroes with images and descriptions using LazyColumn and Card components.

The **30 Days of Fitness Application** presents a daily fitness guide using card layouts containing exercise titles, descriptions, and images.

2 UI Architecture

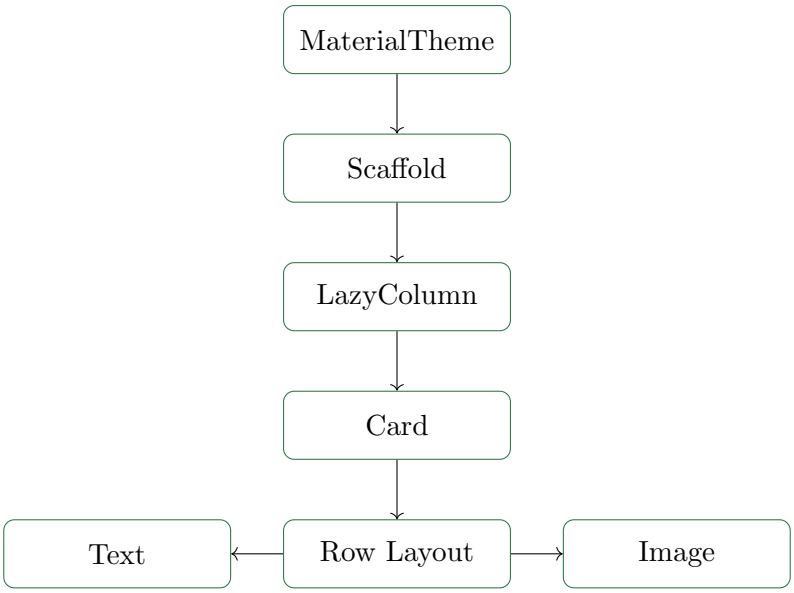

Figure 1: Jetpack Compose UI Architecture

3 Superheroes Application

Light Theme

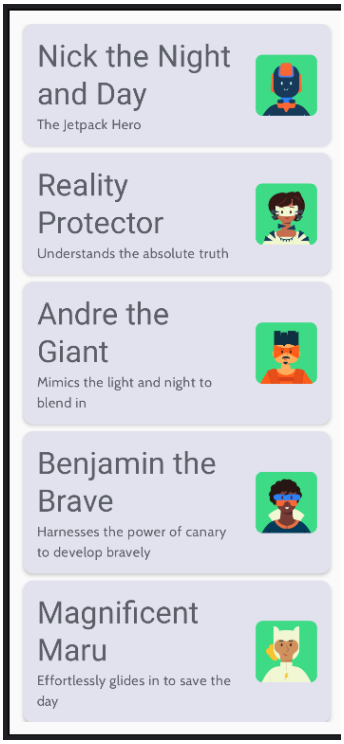

Figure 2: Superheroes Application Light Theme

Dark Theme

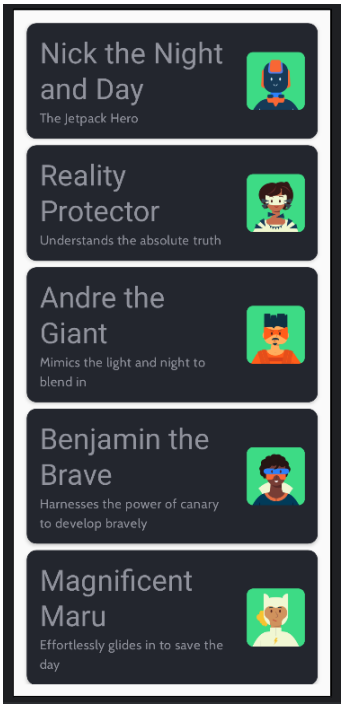

Figure 3: Superheroes Application Dark Theme

The Superheroes application demonstrates the use of LazyColumn for efficient list rendering along with Material 3 card components to display hero information.

4 30 Days of Fitness Application

Light Theme

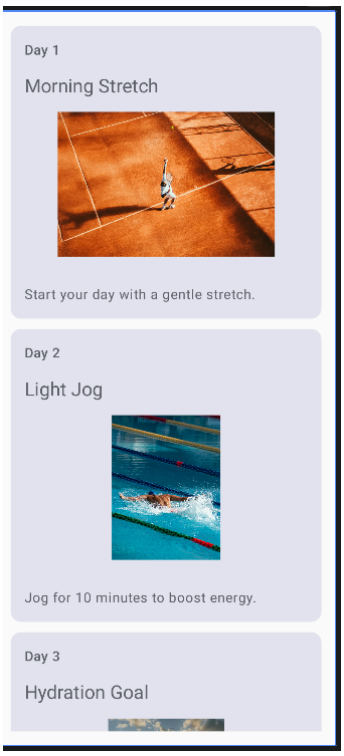

Figure 4: Fitness Application Light Theme

Dark Theme

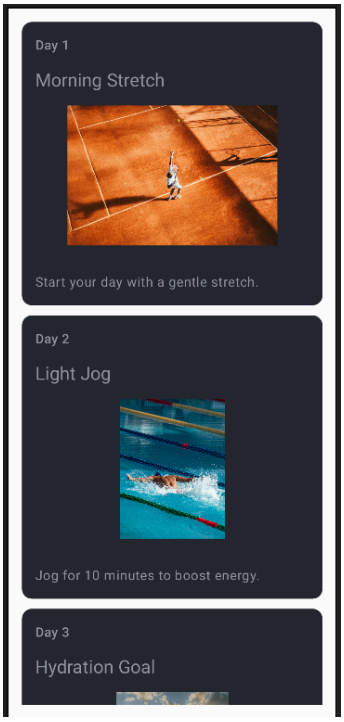

Figure 5: Fitness Application Dark Theme

The Fitness application presents daily exercise activities using card-based UI components. Each card displays an image, title, and description of the workout activity.

5 Implementation Details

Compose Components Used

- Scaffold
- LazyColumn
- Card
- Row Layout
- Column Layout
- Image Composable
- Text Composable

These components allow the applications to maintain a clean UI structure while ensuring responsive layouts across devices.

6 Performance Evaluation

Evaluation Aspect	Observation
UI Responsiveness	Smooth scrolling with LazyColumn
Theme Support	Light and Dark themes render correctly
Layout Adaptability	UI adjusts properly across screen sizes
Performance	No lag observed during rendering

7 Reflection

Developing these applications provided practical experience in building modern Android user interfaces using Jetpack Compose.

Material Design 3 ensures visual consistency across the application while Compose simplifies UI development by using declarative layouts.

The modular UI structure also makes the application easier to maintain and extend with additional features.

8 Future Enhancements

Possible improvements include:

- Adding interactive card animations

- Supporting dynamic color themes
- Adding expandable fitness detail cards
- Optimizing image loading performance